CAVAN DONOHOE

Looking to improve your company's data workflow

Senior Data Scientist / Statistical Programmer with 6+ years of experience building production-grade R packages, automated clinical trial pipelines, and reproducible TFL workflows in regulated environments. Deep expertise in R, CDISC standards (SDTM/ADaM), statistical programming, and end-to-end automation across large-scale clinical datasets. Proven track record of saving thousands of analyst hours through tooling, package development, and workflow optimization.

PROFESSIONAL EXPERIENCE

February 2021 | August 2024

Senior Data Scientist

GRAIL — Menlo Park, CA

- Authored and maintained multiple production R packages used company-wide for clinical trial analysis, validation, and reporting.
- Built automated R Markdown pipelines generating hundreds of Tables, Figures, and Listings (TFLs) to support trial accuracy assessments and regulatory reporting.
- Developed and documented derived variables under CDISC standards (SDTM, ADaM, CDASH) with Git/GitLab collaboration and AWS S3 data management.
- Created an R Shiny app to visualize cancer screening timelines and key events leading to diagnosis.
- Provided statistical programming support for data cleaning, database locks, SAP-driven analyses, and exploratory investigations.
- Partnered with biostatisticians to translate SAPs into analysis specifications and reusable code.
- Built reproducible pipelines with unit tests and a shared code library to ensure audit-ready, backward-compatible outputs.

April 2020 | January 2021

Reporting Analyst

24 Hour Home Care — El Segundo, CA

- Automated weekly accounts receivable aging report in R with HTML/CSS email delivery, saving ~5 hours per week.
- Built ad hoc dashboards and data visualizations to support leadership requests.
- Served as analytics liaison across teams to align reporting needs and delivery.
- Automated batch invoice emails and Azure Web App exports, reducing manual effort across teams.

January 2019 | March 2020

Actuarial Analyst

HMSA — Honolulu, HI

- Automated CRG band renewal in R, reducing runtime from 8 hours to under 1 minute.
- Automated demographic and plan risk factor workflows and forecasting models with Excel/R integration.
- Automated CRG proposal and mental health parity processes; documented methods and led stakeholder walkthroughs.
- Fit empirical distributions using KS tests and MLE for actuarial modeling.

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CONTACT INFO

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SKILLS

R: tidyverse, data.table, ggplot2, Shiny, R Markdown, package development, unit testing, resampling

Clinical & Statistics: CDISC (SDTM, ADaM, CDASH), survival analysis, model validation

Languages & Data: Python, SQL (PostgreSQL), AWS S3

Git & CI/CD: Git, GitHub, GitLab, GitHub Actions (scheduled pipelines, test/lint gates, preview deploys), GitLab CI, Docker

AI-assisted development (Cursor, Copilot, Codex) with full manual review

Excel: Advanced formulas, VBA

SELECTED PROJECTS

Personal Data Site
cavandonohoe.github.io

- Built 45+ GitHub Actions workflows that refresh datasets on a schedule (IMDb, S&P 500, rental markets, box office)
- Shipped 225+ merged PRs and 600+ commits through a CI-gated pull request workflow
- Gated every PR with lint, tests, typo and link checks, and Lighthouse/pa11y accessibility audits
- Set up per-branch preview deploys with automatic pruning, Dependabot automerge, and failure alerts
- Resume rebuilds automatically from a Google Sheet, with live GitHub stats, through a scheduled workflow

Rent vs. Buy
R Shiny app

- Interactive app comparing the long-run financial outcomes of renting vs. buying a home
- CI runs tests, then deploys automatically to shinyapps.io

Confederate Statues
GitHub

- Scraped Wikipedia in R and Python to build a dataset of when Confederate statues were erected
- Data refreshes on a schedule, with CI and a Docker build for the service

EDUCATION

2018

Bachelor of Science
University of California, Irvine — Irvine, CA

- Major in Mathematics
- Minor in Statistics

AWARDS

BAS UCLA Case Competition Finalist (Project Presentation and Project Memorandum)

CREDENTIALS

Passed 3 Actuarial Exams (Exam P, FM, SRM)